

PHASE-I PROJECT PROPOSAL

Digital Trust Score Framework for Cloud Applications using Explainable Artificial Intelligence and Trust Analytics

Course Code: BCSE355L

Course Title: Cloud Architecture Design

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1. Abstract

This Phase-I proposal presents the planning and conceptual design of a Digital Trust Score Framework for Cloud Applications using Explainable Artificial Intelligence (XAI) and Trust Analytics. Users currently depend on fragmented information such as SSL checkers, WHOIS services, reputation websites and online reviews to determine whether a cloud-hosted web application is trustworthy. The proposed framework intends to unify these trust indicators into a single explainable trust score. The system will collect measurable trust features, apply a machine learning model, and use explainability techniques to justify the prediction instead of presenting only a numeric score. AWS services are proposed for deployment, storage, monitoring and security. This document reflects the work completed during Phase-I and serves as the foundation for future implementation.

2. Introduction

Trust is an essential factor in digital interactions. Before sharing personal information or completing online transactions, users often evaluate whether a website appears secure, legitimate and reliable. Existing trust assessment tools provide valuable but isolated information, requiring users to manually combine multiple observations. This project aims to investigate whether multiple trust indicators can be integrated into one intelligent and explainable framework.

3. Problem Statement

Users currently rely on fragmented online services to evaluate trustworthiness. Existing systems frequently lack transparency regarding how trust scores are produced. There is a need for an intelligent framework capable of combining measurable trust indicators into a single explainable assessment.

4. Motivation

During initial project discussions we observed that users naturally consider many signals before trusting a website, including SSL certificates, domain history, contact information, company legitimacy, reviews and technical security. Rather than checking these independently, our framework proposes to aggregate them into one decision-support system.

5. Project Objectives

- Develop a trust evaluation framework for cloud-hosted web applications.
- Identify measurable trust indicators from public sources.

- Investigate ML-based trust prediction.
- Provide explainable AI reasoning for each prediction.
- Deploy the solution using AWS cloud services.
- Design a dashboard for trust analytics.

6. Project Scope

In Scope

- Website URL analysis
- Collection of observable trust indicators
- Explainable trust score
- Dashboard visualization

Out of Scope

- Penetration testing
- Zero-day vulnerability detection
- Dark web monitoring
- Guaranteeing website safety

7. Existing Systems

Initial study of publicly available trust assessment platforms indicates that most systems focus on one or more of the following: SSL validation, domain information, malware reputation, website popularity or user reviews. While these systems provide useful information, the proposed framework intends to emphasize transparency by explaining how each collected feature contributes to the final trust score.

8. Proposed Framework

Workflow:

User enters Website URL

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Feature Collection

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Trust Indicator Extraction

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Machine Learning Prediction

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Explainable AI Module

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Digital Trust Score
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Trust Analytics Dashboard

9. Initial Trust Feature Specification

Category	Feature	Purpose	Status	Difficulty
Security	HTTPS	Encrypted communication	Candidate	Easy
Security	SSL Certificate	Certificate validity	Candidate	Easy
Security	Security Headers	Secure configuration	Candidate	Easy
Reputation	Domain Age	Domain credibility	Candidate	Easy
Reputation	Blacklist Status	Known malicious reports	Candidate	Medium
Transparency	Privacy Policy	User transparency	Candidate	Easy
Transparency	Contact Page	Business legitimacy	Candidate	Easy
Reliability	Response Time	Performance indicator	Candidate	Easy

10. Preliminary AWS Architecture

Proposed mapping:

- Amazon EC2 – Backend hosting
- Amazon S3 – Datasets and reports
- Amazon RDS – Store analysis history
- AWS IAM – Authentication and authorization
- Amazon CloudWatch – Monitoring
- Amazon SNS – Future notifications

Architecture diagram will be finalized after feature validation.

11. Technology Stack

Component	Technology
Frontend	React (Tentative)
Backend	Python FastAPI/Flask

ML	Random Forest/XGBoost (Under Evaluation)
Explainability	SHAP
Cloud	AWS
Version Control	Git & GitHub

12. Literature Survey (In Progress)

A comprehensive literature survey has been planned in accordance with the course guidelines. The initial review has identified the following research domains:

- Explainable Artificial Intelligence
- Website Trust Evaluation
- Trust Analytics
- Phishing Detection
- Cloud Security
- Machine Learning for Cybersecurity

Detailed analysis of fifteen research papers will be completed during the next phase.

13. Preliminary Research Gap

Based on the initial investigation, current tools provide fragmented trust information and limited explainability. The proposed work aims to investigate whether integrating multiple trust dimensions with explainable AI can improve transparency and user understanding. This research gap will be refined after the literature survey.

14. Dataset Strategy

Suitable public datasets and feature sources are under evaluation. The project may combine publicly available datasets with automatically extracted website trust indicators, depending on feasibility and literature findings.

15. Work Distribution

Individual Project

Krishna Garg (24BIT0209): Complete project planning, literature survey, research gap analysis, architecture design, implementation, AWS deployment, documentation and testing.

16. Project Timeline

- Phase-I: Planning and proposal

- Literature survey
- Architecture design
- Dataset selection
- Implementation
- Testing
- Deployment
- Final documentation

17. Future Work

The next project phase will focus on validating trust indicators through research papers, finalizing the architecture, selecting datasets, implementing feature extraction modules, training an explainable machine learning model and deploying the solution on AWS.

18. References

1. AWS Documentation.
2. Research papers (to be added after literature review).